

*Answer key***Problem 1.**

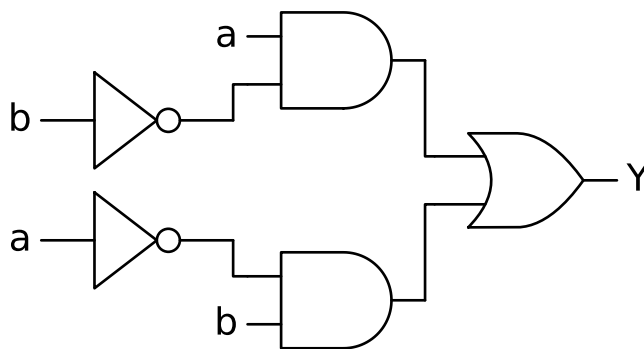
From the truth table below, write the Boolean expression for output Y as a sum of products (one product term for each row where $Y = 1$), then draw its logic-gate circuit.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Expression: $A \cdot \overline{B} + \overline{A} \cdot B$

Simplifies to: $A \cdot \overline{B} + B \cdot \overline{A}$

Circuit:

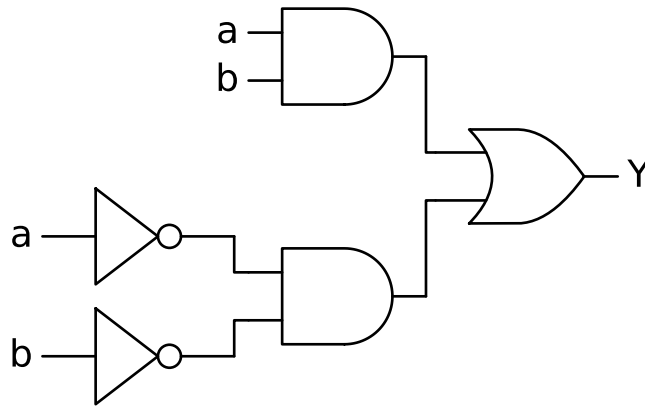
**Problem 2.**

From the truth table below, write the Boolean expression for output Y as a sum of products (one product term for each row where $Y = 1$), then draw its logic-gate circuit.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Expression: $A \cdot B + \overline{A} \cdot \overline{B}$

Circuit:



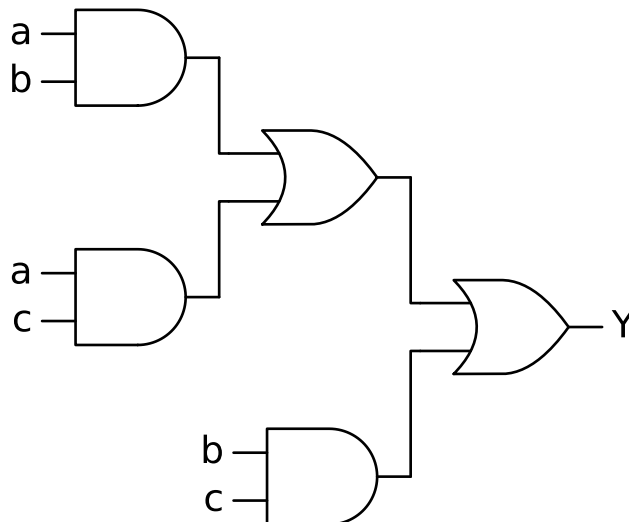
Problem 3.

From the truth table below, write the Boolean expression for output Y as a sum of products (one product term for each row where $Y = 1$), then draw its logic-gate circuit.

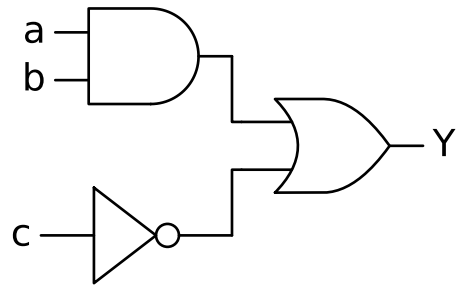
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Expression: $A \cdot B + A \cdot C + B \cdot C$

Circuit:



Problem 4.



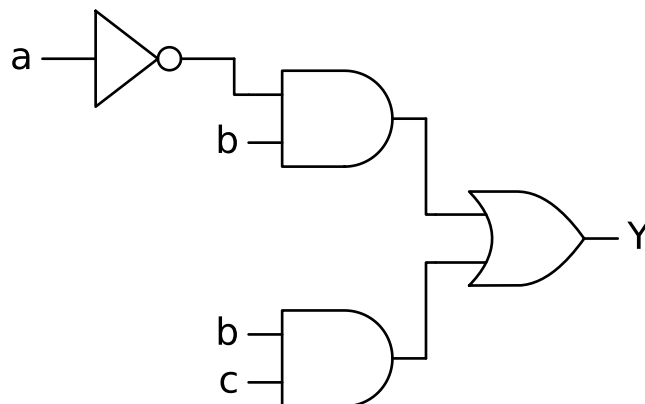
Expression: $A \cdot B + \overline{C}$

Simplifies to: $\overline{C} + A \cdot B$

Truth table:

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Problem 5.



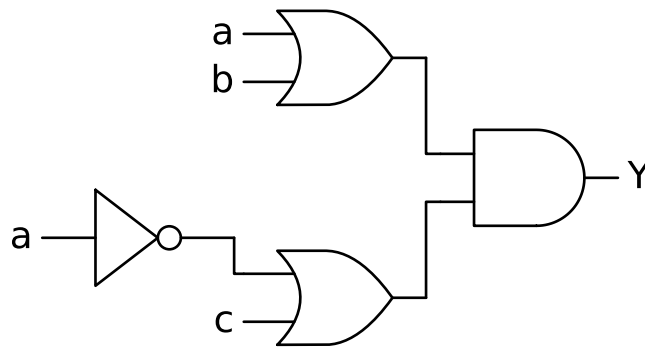
Expression: $\overline{A} \cdot B + B \cdot C$

Simplifies to: $B \cdot C + B \cdot \overline{A}$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Problem 6.



Expression: $(A + B) \cdot (\bar{A} + C)$

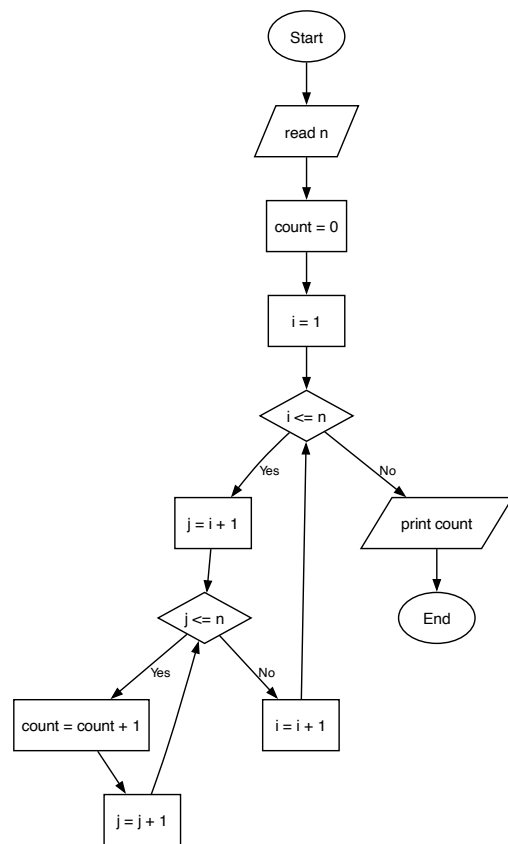
Simplifies to: $A \cdot C + B \cdot \bar{A}$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Problem 7.

Trace this algorithm for $n = 3$, and complete the trace table.



n	count	i	j	output
3				
3	0			
3	0	1		
3	0	1	2	
3	1	1	3	
3	2	1	3	
3	2	1	4	
3	2	2	4	
3	2	2	3	
3	3	2	3	
3	3	2	4	
3	3	3	4	
3	3	3	4	
3	3	4	4	
3	3	4	4	3

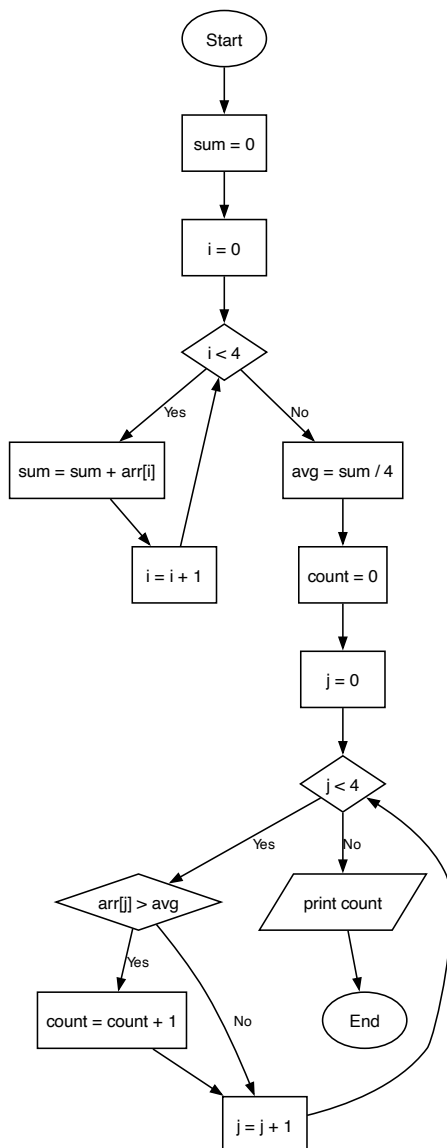
Output: 3

In the real world this is how you count the number of distinct pairs in a group of n things: the number of handshakes if n people all shake hands once, or the number of direct links in a network where every one of n computers connects to every other. It always works out to $n \times (n - 1) / 2$.

Problem 8.

Trace this algorithm, taking the list *arr* to be [70, 90, 60, 80], and complete the trace table.

Reminder: the items of a list have positions counted from 0, so the list [60, 75, 50] has 3 items in positions 0, 1, 2 — $arr[i]$ means the item at position i , so $arr[0]$ is the first item.



sum	i	avg	count	j	output
0					
0	0				
70	0				
70	1				
160	1				
160	2				
220	2				
220	3				
300	3				
300	4				
300	4	75			
300	4	75	0		
300	4	75	0	0	
300	4	75	0	1	
300	4	75	1	1	
300	4	75	1	2	
300	4	75	1	3	
300	4	75	2	3	
300	4	75	2	4	
300	4	75	2	4	2

Output: 2

This two-pass pattern — go through the data once to measure something overall, then go through it again to compare each item against that measure — is everywhere in data analysis: for example, finding how many students scored above the class average, or how many days this month were hotter than usual.